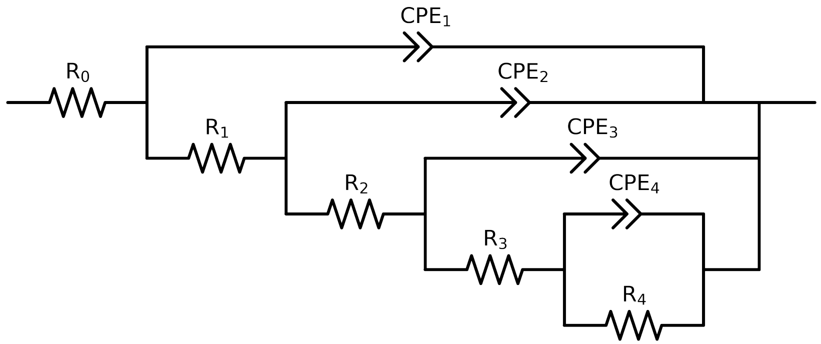
Comparative EIS Kinetics Report

Model Structure: `R0-p(R1-p(R2-p(R3-p(R4,CPE4),CPE3),CPE2),CPE1)`
Used data: original data



Element	**Unit**	**Bounds**	**Vasco_LNV3_4**	**Vasco_LNV3_5**	**Vasco_LNV3_7d_**
R_0	Ω	$[1.0e+00, 1.0e+03]$	$1.49e+02 \pm 1.4e+00$	$1.25e+02 \pm 2.9e+00$	$1.15e+02 \pm 1.3e+00$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$1.99e+02 \pm 4.9e+01$	$8.28e+02 \pm 8.1e+01$	$4.69e+02 \pm 4.4e+01$
R_2	Ω	$[1.0e+00, 1.0e+08]$	$6.08e+02 \pm 2.0e+02$	$4.19e+05 \pm 1.6e+05$	$1.47e+03 \pm 2.2e+02$
R_3	Ω	$[1.0e+00, 1.0e+08]$	$1.91e+05 \pm 1.6e+04$	$1.65e+05 \pm 1.7e+05$	$2.43e+05 \pm 8.6e+03$
R_4	Ω	$[1.0e+00, 1.0e+08]$	$1.28e+06 \pm 1.6e+05$	$4.36e+05 \pm 1.7e+05$	$6.50e+05 \pm 7.0e+04$
Q_4	Ω^{-1} sec^a	$[1.0e-09, 1.0e-03]$	$1.24e-05 \pm 7.4e-07$	$6.39e-08 \pm 3.6e-02$	$4.15e-05 \pm 3.5e-06$
n_4	-	$[5.0e-01, 1.0e+00]$	$6.97e-01 \pm 3.0e-02$	$1.00e+00 \pm 2.3e+05$	$9.43e-01 \pm 3.3e-02$
Q_3	Ω^{-1} sec^a	$[1.0e-09, 1.0e-03]$	$2.97e-06 \pm 1.2e-06$	$4.54e-05 \pm 1.9e-02$	$5.80e-06 \pm 1.0e-06$
n_3	-	$[5.0e-01, 1.0e+00]$	$9.12e-01 \pm 3.7e-02$	$1.00e+00 \pm 1.7e+02$	$8.76e-01 \pm 2.1e-02$
Q_2	Ω^{-1} sec^a	$[1.0e-09, 1.0e-02]$	$1.70e-06 \pm 1.9e-06$	$5.96e-06 \pm 9.2e-07$	$3.80e-06 \pm 1.5e-06$
n_2	-	$[5.0e-01, 1.0e+00]$	$9.06e-01 \pm 1.3e-01$	$8.75e-01 \pm 2.1e-02$	$8.27e-01 \pm 5.1e-02$
Q_1	Ω^{-1} sec^a	$[1.0e-09, 1.0e-03]$	$2.00e-06 \pm 1.1e-06$	$5.85e-06 \pm 1.4e-06$	$3.30e-06 \pm 6.6e-07$
n_1	-	$[5.0e-01, 1.0e+00]$	$7.92e-01 \pm 4.4e-02$	$6.87e-01 \pm 2.4e-02$	$7.04e-01 \pm 1.7e-02$
χ^2	-	-	$1.295e-04$	$1.513e-03$	$9.282e-05$

Analysis Results: Vasco_LNV3_4

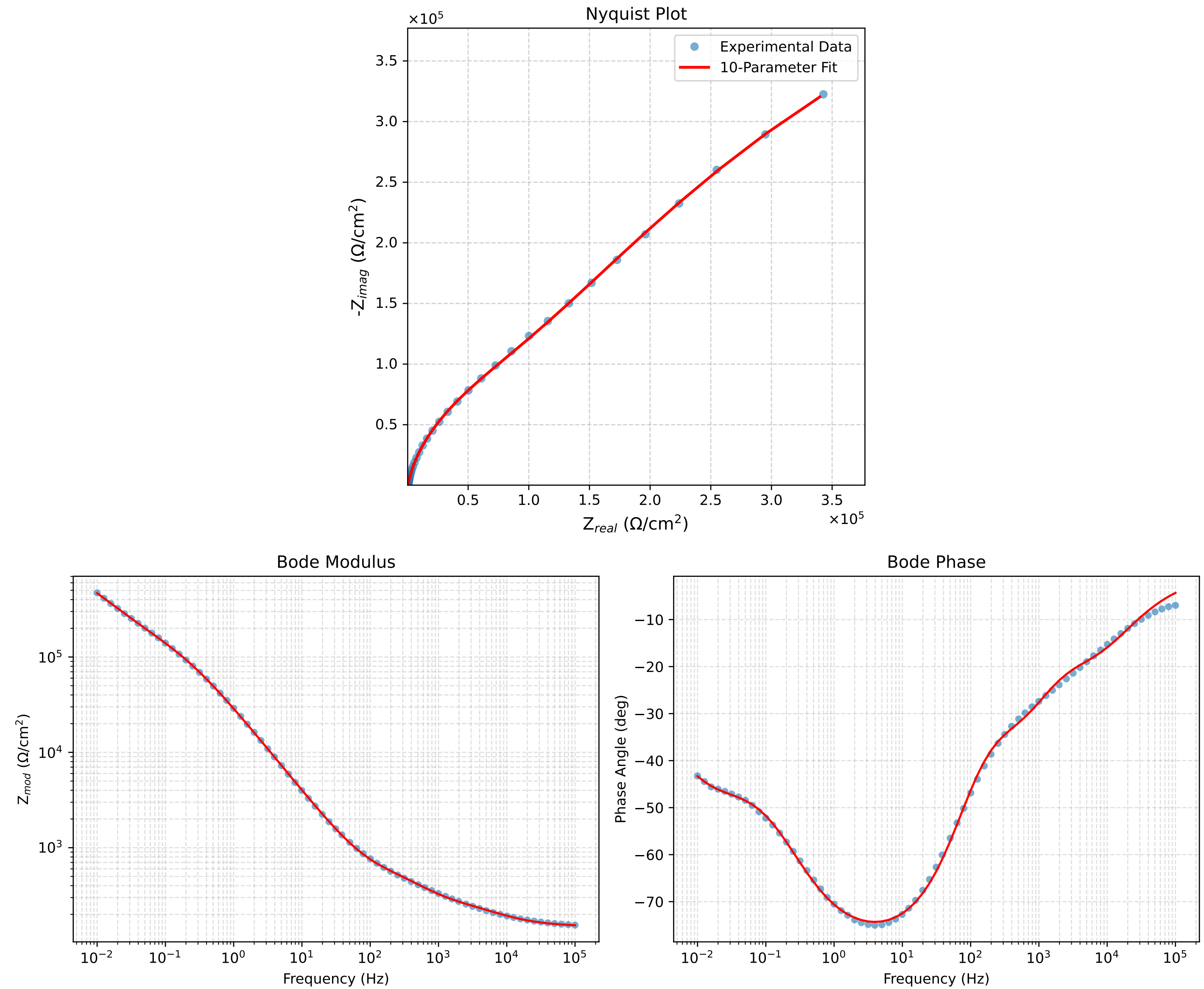


Figure 1: EIS Fit Results for Vasco_LNV3_4

Fitted Data: Vasco_LNV3_4

Frequency (Hz)	Z_{real} (Ω)	$-Z_{imag}$ (Ω)	Z_{mod} (Ω)	Z_{phase} (°)
1.0008e+05	1.5335e+02	1.1572e+01	1.5379e+02	-4.3152
7.9453e+04	1.5442e+02	1.3757e+01	1.5503e+02	-5.0907
6.3141e+04	1.5574e+02	1.6305e+01	1.5660e+02	-5.9764
5.0203e+04	1.5739e+02	1.9262e+01	1.5856e+02	-6.9775
3.9891e+04	1.5944e+02	2.2686e+01	1.6105e+02	-8.0977
3.1641e+04	1.6202e+02	2.6631e+01	1.6419e+02	-9.3342
2.5172e+04	1.6517e+02	3.1028e+01	1.6806e+02	-10.6395
2.0016e+04	1.6905e+02	3.5926e+01	1.7282e+02	-11.9979
1.5891e+04	1.7379e+02	4.1309e+01	1.7864e+02	-13.3705
1.2609e+04	1.7948e+02	4.7078e+01	1.8555e+02	-14.6976
1.0078e+04	1.8592e+02	5.2924e+01	1.9330e+02	-15.8898
8.0156e+03	1.9340e+02	5.9059e+01	2.0221e+02	-16.9816
6.3281e+03	2.0194e+02	6.5505e+01	2.1230e+02	-17.9717
5.0156e+03	2.1097e+02	7.2007e+01	2.2292e+02	-18.8457
3.9844e+03	2.2030e+02	7.8812e+01	2.3398e+02	-19.6843
3.1710e+03	2.2983e+02	8.6284e+01	2.4550e+02	-20.5770
2.5276e+03	2.3958e+02	9.4883e+01	2.5769e+02	-21.6050
1.9761e+03	2.5074e+02	1.0613e+02	2.7227e+02	-22.9407
1.5775e+03	2.6189e+02	1.1863e+02	2.8751e+02	-24.3683
1.2656e+03	2.7420e+02	1.3318e+02	3.0483e+02	-25.9057
9.9826e+02	2.8961e+02	1.5153e+02	3.2685e+02	-27.6193
7.9688e+02	3.0677e+02	1.7139e+02	3.5140e+02	-29.1922
6.2779e+02	3.2784e+02	1.9473e+02	3.8131e+02	-30.7095
5.0551e+02	3.4936e+02	2.1778e+02	4.1168e+02	-31.9386
3.9800e+02	3.7505e+02	2.4541e+02	4.4821e+02	-33.1987
3.1550e+02	4.0093e+02	2.7524e+02	4.8632e+02	-34.4695
2.5240e+02	4.2571e+02	3.0821e+02	5.2557e+02	-35.9040
1.9862e+02	4.5152e+02	3.5087e+02	5.7182e+02	-37.8498
1.5836e+02	4.7504e+02	4.0115e+02	6.2176e+02	-40.1796
1.2556e+02	4.9869e+02	4.6630e+02	6.8273e+02	-43.0778
1.0045e+02	5.2183e+02	5.4565e+02	7.5501e+02	-46.2783
7.9003e+01	5.4847e+02	6.5397e+02	8.5352e+02	-50.0141
6.3345e+01	5.7596e+02	7.7935e+02	9.6908e+02	-53.5345
5.0223e+01	6.0962e+02	9.4353e+02	1.1233e+03	-57.1333
3.8422e+01	6.5712e+02	1.1836e+03	1.3538e+03	-60.9623
3.1250e+01	7.0231e+02	1.4141e+03	1.5789e+03	-63.5885
2.4934e+01	7.6286e+02	1.7208e+03	1.8823e+03	-66.0916
1.9862e+01	8.3888e+02	2.0995e+03	2.2609e+03	-68.2200
1.5625e+01	9.4022e+02	2.5916e+03	2.7568e+03	-70.0592
1.2401e+01	1.0643e+03	3.1749e+03	3.3486e+03	-71.4681
9.9311e+00	1.2151e+03	3.8585e+03	4.0453e+03	-72.5205
7.9449e+00	1.4067e+03	4.6920e+03	4.8983e+03	-73.3109
6.3174e+00	1.6578e+03	5.7329e+03	5.9677e+03	-73.8718
5.0080e+00	1.9853e+03	7.0177e+03	7.2931e+03	-74.2038
3.9457e+00	2.4225e+03	8.6270e+03	8.9607e+03	-74.3148
3.1587e+00	2.9525e+03	1.0447e+04	1.0856e+04	-74.2181
2.5040e+00	3.6724e+03	1.2736e+04	1.3255e+04	-73.9149
1.9981e+00	4.5870e+03	1.5408e+04	1.6076e+04	-73.4212
1.5847e+00	5.8180e+03	1.8685e+04	1.9570e+04	-72.7048
1.2669e+00	7.3744e+03	2.2436e+04	2.3616e+04	-71.8046
9.9904e-01	9.5412e+03	2.7120e+04	2.8749e+04	-70.6172
7.9234e-01	1.2312e+04	3.2451e+04	3.4708e+04	-69.2231
6.3345e-01	1.5758e+04	3.8342e+04	4.1454e+04	-67.6587
5.0403e-01	2.0223e+04	4.5111e+04	4.9436e+04	-65.8535
4.0064e-01	2.5831e+04	5.2631e+04	5.8628e+04	-63.8586
3.1672e-01	3.2869e+04	6.0991e+04	6.9284e+04	-61.6792
2.5202e-01	4.1036e+04	6.9657e+04	8.0846e+04	-59.4971
2.0032e-01	5.0545e+04	7.8813e+04	9.3628e+04	-57.3265
1.5890e-01	6.1393e+04	8.8491e+04	1.0770e+05	-55.2478
1.2601e-01	7.3412e+04	9.8710e+04	1.2302e+05	-53.3614
1.0016e-01	8.6378e+04	1.0955e+05	1.3951e+05	-51.7454
7.9449e-02	1.0055e+05	1.2153e+05	1.5773e+05	-50.3967
6.3174e-02	1.1578e+05	1.3477e+05	1.7767e+05	-49.3349
5.0134e-02	1.3262e+05	1.4990e+05	2.0015e+05	-48.5011
3.9826e-02	1.5127e+05	1.6710e+05	2.2540e+05	-47.8458
3.1630e-02	1.7236e+05	1.8671e+05	2.5411e+05	-47.2882
2.5121e-02	1.9653e+05	2.0890e+05	2.8682e+05	-46.7470
1.9955e-02	2.2452e+05	2.3366e+05	3.2404e+05	-46.1433
1.5852e-02	2.5716e+05	2.6083e+05	3.6628e+05	-45.4055
1.2591e-02	2.9535e+05	2.8998e+05	4.1391e+05	-44.4740
1.0001e-02	3.3988e+05	3.2034e+05	4.6705e+05	-43.3053

Analysis Results: Vasco_LNV3_5

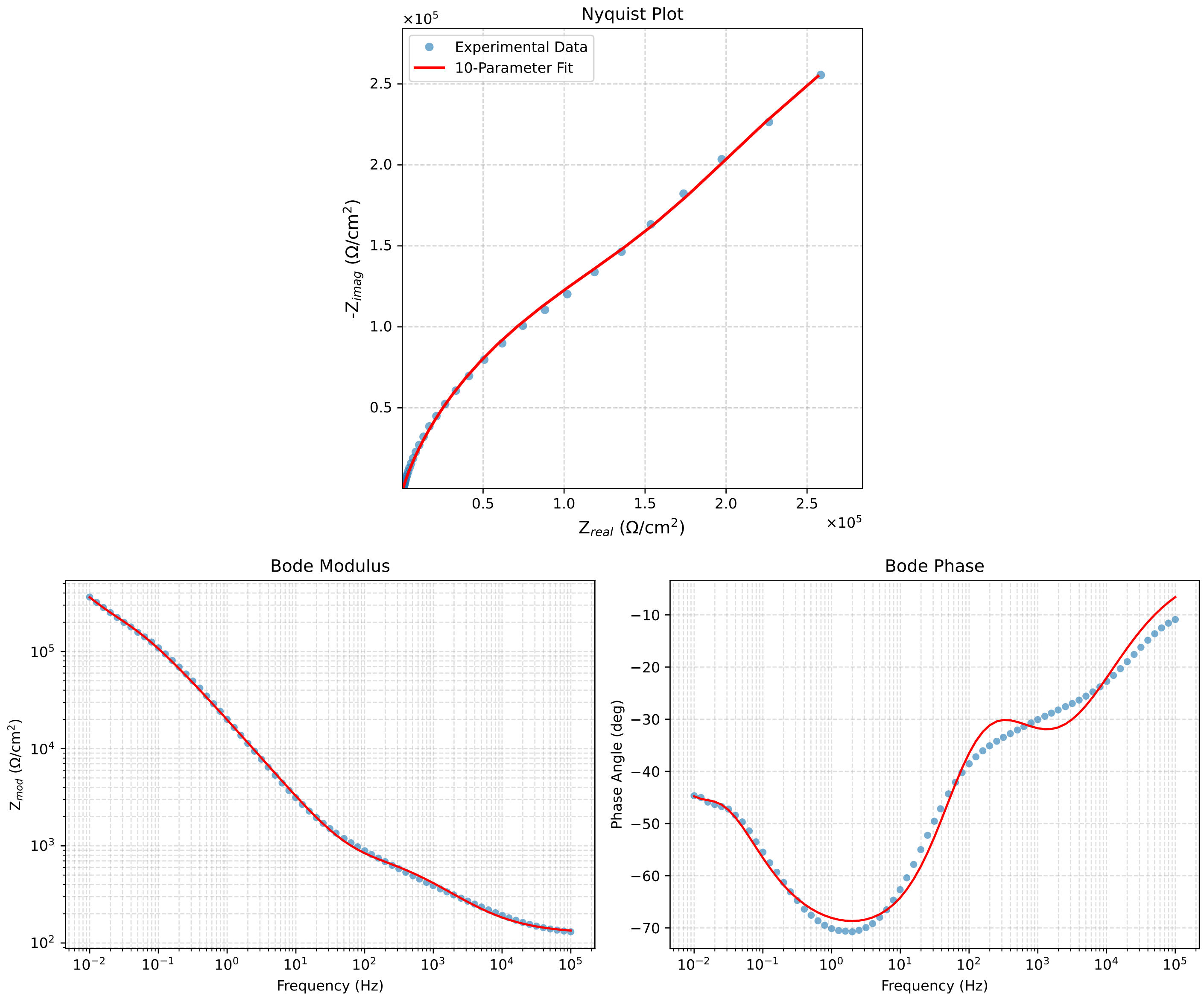


Figure 2: EIS Fit Results for Vasco_LNV3_5

Fitted Data: Vasco_LNV3_5

Frequency (Hz)	Z_{real} (Ω)	-Z_{imag} (Ω)	Z_{mod} (Ω)	Z_{phase} ($^\circ$)
1.0008e+05	1.3338e+02	1.5402e+01	1.3427e+02	-6.5870
7.9453e+04	1.3490e+02	1.7982e+01	1.3609e+02	-7.5926
6.3141e+04	1.3669e+02	2.0967e+01	1.3829e+02	-8.7208
5.0203e+04	1.3880e+02	2.4422e+01	1.4093e+02	-9.9792
3.9891e+04	1.4129e+02	2.8432e+01	1.4413e+02	-11.3776
3.1641e+04	1.4427e+02	3.3105e+01	1.4802e+02	-12.9235
2.5172e+04	1.4776e+02	3.8423e+01	1.5267e+02	-14.5764
2.0016e+04	1.5190e+02	4.4545e+01	1.5830e+02	-16.3441
1.5891e+04	1.5685e+02	5.1605e+01	1.6512e+02	-18.2120
1.2609e+04	1.6274e+02	5.9677e+01	1.7334e+02	-20.1382
1.0078e+04	1.6951e+02	6.8538e+01	1.8284e+02	-22.0150
8.0156e+03	1.7770e+02	7.8728e+01	1.9436e+02	-23.8949
6.3281e+03	1.8775e+02	9.0514e+01	2.0843e+02	-25.7381
5.0156e+03	1.9949e+02	1.0339e+02	2.2469e+02	-27.3961
3.9844e+03	2.1320e+02	1.1737e+02	2.4338e+02	-28.8339
3.1710e+03	2.2915e+02	1.3238e+02	2.6464e+02	-30.0159
2.5276e+03	2.4755e+02	1.4826e+02	2.8855e+02	-30.9178
1.9761e+03	2.7065e+02	1.6632e+02	3.1767e+02	-31.5715
1.5775e+03	2.9477e+02	1.8330e+02	3.4711e+02	-31.8746
1.2656e+03	3.2107e+02	1.9999e+02	3.7826e+02	-31.9180
9.9826e+02	3.5217e+02	2.1774e+02	4.1404e+02	-31.7277
7.9688e+02	3.8391e+02	2.3416e+02	4.4969e+02	-31.3810
6.2779e+02	4.1916e+02	2.5111e+02	4.8862e+02	-30.9253
5.0551e+02	4.5191e+02	2.6644e+02	5.2461e+02	-30.5231
3.9800e+02	4.8803e+02	2.8411e+02	5.6470e+02	-30.2062
3.1550e+02	5.2231e+02	3.0331e+02	6.0399e+02	-30.1439
2.5240e+02	5.5403e+02	3.2529e+02	6.4247e+02	-30.4181
1.9862e+02	5.8663e+02	3.5496e+02	6.8566e+02	-31.1776
1.5836e+02	6.1626e+02	3.9121e+02	7.2995e+02	-32.4079
1.2556e+02	6.4606e+02	4.3937e+02	7.8131e+02	-34.2185
1.0045e+02	6.7506e+02	4.9898e+02	8.3946e+02	-36.4703
7.9003e+01	7.0797e+02	5.8118e+02	9.1596e+02	-39.3831
6.3345e+01	7.4122e+02	6.7682e+02	1.0037e+03	-42.3997
5.0223e+01	7.8100e+02	8.0218e+02	1.1196e+03	-45.7667
3.8422e+01	8.3576e+02	9.8502e+02	1.2918e+03	-49.6864
3.1250e+01	8.8674e+02	1.1595e+03	1.4597e+03	-52.5931
2.4934e+01	9.5375e+02	1.3900e+03	1.6858e+03	-55.5444
1.9862e+01	1.0362e+03	1.6719e+03	1.9669e+03	-58.2091
1.5625e+01	1.1437e+03	2.0338e+03	2.3334e+03	-60.6484
1.2401e+01	1.2721e+03	2.4574e+03	2.7671e+03	-62.6312
9.9311e+00	1.4239e+03	2.9472e+03	3.2731e+03	-64.2132
7.9449e+00	1.6109e+03	3.5363e+03	3.8859e+03	-65.5090
6.3174e+00	1.8473e+03	4.2618e+03	4.6449e+03	-66.5650
5.0080e+00	2.1433e+03	5.1448e+03	5.5734e+03	-67.3834
3.9457e+00	2.5203e+03	6.2357e+03	6.7258e+03	-67.9926
3.1587e+00	2.9550e+03	7.4542e+03	8.0185e+03	-68.3757
2.5040e+00	3.5146e+03	8.9711e+03	9.6350e+03	-68.6061
1.9981e+00	4.1867e+03	1.0728e+04	1.1516e+04	-68.6803
1.5847e+00	5.0408e+03	1.2873e+04	1.3824e+04	-68.6151
1.2669e+00	6.0613e+03	1.5328e+04	1.6483e+04	-68.4236
9.9904e-01	7.4077e+03	1.8414e+04	1.9848e+04	-68.0860
7.9234e-01	9.0494e+03	2.1981e+04	2.3771e+04	-67.6232
6.3345e-01	1.1020e+04	2.6021e+04	2.8258e+04	-67.0478
5.0403e-01	1.3519e+04	3.0832e+04	3.3665e+04	-66.3238
4.0064e-01	1.6648e+04	3.6444e+04	4.0067e+04	-65.4489
3.1672e-01	2.0650e+04	4.3079e+04	4.7772e+04	-64.3892
2.5202e-01	2.5496e+04	5.0450e+04	5.6527e+04	-63.1893
2.0032e-01	3.1518e+04	5.8798e+04	6.6712e+04	-61.8072
1.5890e-01	3.8987e+04	6.8158e+04	7.8521e+04	-60.2300
1.2601e-01	4.8085e+04	7.8389e+04	9.1962e+04	-58.4741
1.0016e-01	5.8859e+04	8.9212e+04	1.0688e+05	-56.5844
7.9449e-02	7.1590e+04	1.0063e+05	1.2350e+05	-54.5707
6.3174e-02	8.5957e+04	1.1221e+05	1.4135e+05	-52.5464
5.0134e-02	1.0199e+05	1.2407e+05	1.6061e+05	-50.5786
3.9826e-02	1.1913e+05	1.3618e+05	1.8094e+05	-48.8219
3.1630e-02	1.3712e+05	1.4909e+05	2.0255e+05	-47.3950
2.5121e-02	1.5585e+05	1.6360e+05	2.2595e+05	-46.3894
1.9955e-02	1.7574e+05	1.8075e+05	2.5211e+05	-45.8053
1.5852e-02	1.9793e+05	2.0152e+05	2.8246e+05	-45.5155
1.2591e-02	2.2421e+05	2.2637e+05	3.1861e+05	-45.2749
1.0001e-02	2.5674e+05	2.5479e+05	3.6171e+05	-44.7814

Analysis Results: Vasco_LNV3_7d_

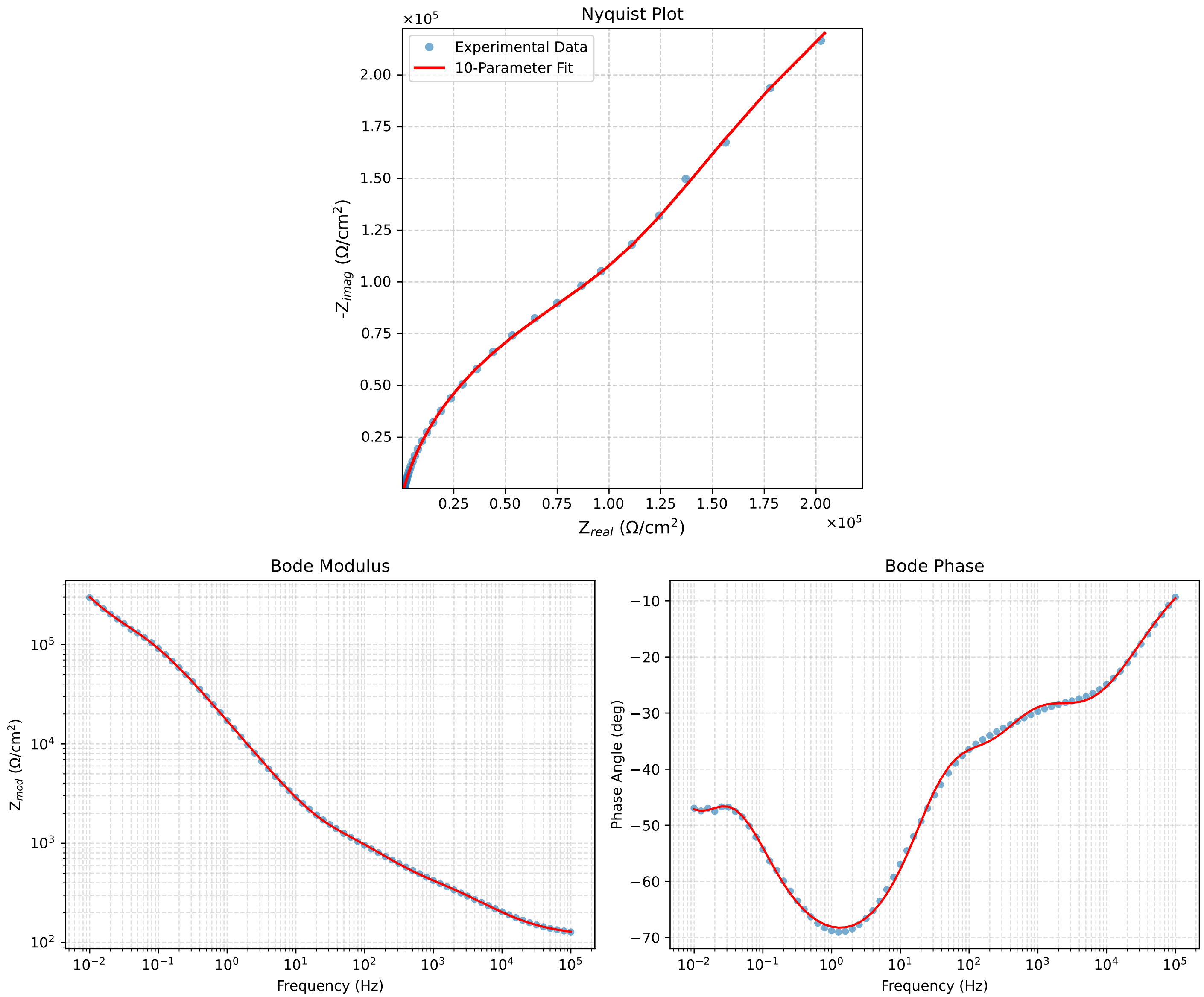


Figure 3: EIS Fit Results for Vasco_LNV3_7d_

Fitted Data: Vasco_LNV3_7d_

Frequency (Hz)	$\mathbf{Z}_{real}$ (Ω)	-$\mathbf{Z}_{imag}$ (Ω)	$\mathbf{Z}_{mod}$ (Ω)	$\mathbf{Z}_{phase}$ ($^\circ$)
1.0002e+05	1.2716e+02	2.1345e+01	1.2894e+02	-9.5288
7.9512e+04	1.2939e+02	2.4861e+01	1.3175e+02	-10.8766
6.3105e+04	1.3207e+02	2.8939e+01	1.3520e+02	-12.3594
5.0215e+04	1.3524e+02	3.3560e+01	1.3934e+02	-13.9370
3.9902e+04	1.3904e+02	3.8860e+01	1.4437e+02	-15.6147
3.1699e+04	1.4360e+02	4.4874e+01	1.5044e+02	-17.3540
2.5137e+04	1.4908e+02	5.1697e+01	1.5779e+02	-19.1250
1.9980e+04	1.5555e+02	5.9229e+01	1.6645e+02	-20.8451
1.5879e+04	1.6323e+02	6.7541e+01	1.7665e+02	-22.4785
1.2598e+04	1.7237e+02	7.6664e+01	1.8865e+02	-23.9782
1.0020e+04	1.8294e+02	8.6348e+01	2.0230e+02	-25.2673
7.9282e+03	1.9548e+02	9.6803e+01	2.1813e+02	-26.3456
6.3079e+03	2.0947e+02	1.0740e+02	2.3540e+02	-27.1441
5.0347e+03	2.2497e+02	1.1805e+02	2.5406e+02	-27.6869
3.9931e+03	2.4256e+02	1.2907e+02	2.7476e+02	-28.0177
3.1441e+03	2.6222e+02	1.4043e+02	2.9746e+02	-28.1711
2.5195e+03	2.8155e+02	1.5103e+02	3.1950e+02	-28.2106
2.0008e+03	3.0252e+02	1.6240e+02	3.4336e+02	-28.2281
1.5807e+03	3.2462e+02	1.7485e+02	3.6872e+02	-28.3089
1.2536e+03	3.4692e+02	1.8859e+02	3.9487e+02	-28.5286
1.0007e+03	3.6929e+02	2.0410e+02	4.2194e+02	-28.9281
7.9003e+02	3.9393e+02	2.2348e+02	4.5290e+02	-29.5665
6.2881e+02	4.1952e+02	2.4593e+02	4.8629e+02	-30.3794
5.0223e+02	4.4732e+02	2.7216e+02	5.2361e+02	-31.3171
4.0015e+02	4.7900e+02	3.0311e+02	5.6685e+02	-32.3254
3.1550e+02	5.1701e+02	3.4025e+02	6.1892e+02	-33.3495
2.5202e+02	5.5827e+02	3.7955e+02	6.7507e+02	-34.2109
2.0032e+02	6.0612e+02	4.2347e+02	7.3940e+02	-34.9401
1.5801e+02	6.6135e+02	4.7245e+02	8.1277e+02	-35.5412
1.2556e+02	7.1946e+02	5.2349e+02	8.8976e+02	-36.0401
9.9734e+01	7.8073e+02	5.7917e+02	9.7210e+02	-36.5692
7.9449e+01	8.4263e+02	6.4086e+02	1.0586e+03	-37.2547
6.2919e+01	9.0623e+02	7.1477e+02	1.1542e+03	-38.2640
4.9867e+01	9.6916e+02	8.0420e+02	1.2594e+03	-39.6857
3.8422e+01	1.0398e+03	9.3081e+02	1.3956e+03	-41.8343
3.1250e+01	1.0972e+03	1.0571e+03	1.5236e+03	-43.9334
2.4934e+01	1.1637e+03	1.2283e+03	1.6920e+03	-46.5478
2.0032e+01	1.2345e+03	1.4344e+03	1.8925e+03	-49.2844
1.5625e+01	1.3267e+03	1.7266e+03	2.1774e+03	-52.4622
1.2467e+01	1.4258e+03	2.0566e+03	2.5025e+03	-55.2669
9.9734e+00	1.5431e+03	2.4550e+03	2.8997e+03	-57.8481
7.9719e+00	1.6860e+03	2.9411e+03	3.3901e+03	-60.1757
6.3516e+00	1.8638e+03	3.5389e+03	3.9997e+03	-62.2266
5.0134e+00	2.0935e+03	4.2954e+03	4.7784e+03	-64.0163
3.9860e+00	2.3713e+03	5.1843e+03	5.7009e+03	-65.4206
3.1758e+00	2.7145e+03	6.2453e+03	6.8097e+03	-66.5082
2.5161e+00	3.1550e+03	7.5538e+03	8.1863e+03	-67.3310
1.9955e+00	3.7074e+03	9.1203e+03	9.8450e+03	-67.8785
1.5879e+00	4.3936e+03	1.0969e+04	1.1816e+04	-68.1709
1.2581e+00	5.2782e+03	1.3217e+04	1.4232e+04	-68.2302
9.9777e-01	6.3973e+03	1.5880e+04	1.7121e+04	-68.0584
7.9274e-01	7.8097e+03	1.9007e+04	2.0549e+04	-67.6633
6.2903e-01	9.6189e+03	2.2703e+04	2.4657e+04	-67.0385
4.9888e-01	1.1931e+04	2.7023e+04	2.9540e+04	-66.1783
3.9860e-01	1.4769e+04	3.1841e+04	3.5099e+04	-65.1156
3.1672e-01	1.8439e+04	3.7451e+04	4.1744e+04	-63.7865
2.5148e-01	2.3061e+04	4.3755e+04	4.9460e+04	-62.2079
1.9998e-01	2.8750e+04	5.0625e+04	5.8219e+04	-60.4073
1.5879e-01	3.5715e+04	5.8031e+04	6.8141e+04	-58.3900
1.2601e-01	4.4009e+04	6.5782e+04	7.9146e+04	-56.2171
1.0016e-01	5.3486e+04	7.3617e+04	9.0996e+04	-53.9999
7.9503e-02	6.4077e+04	8.1525e+04	1.0369e+05	-51.8333
6.3140e-02	7.5389e+04	8.9483e+04	1.1701e+05	-49.8861
5.0123e-02	8.7137e+04	9.7816e+04	1.3100e+05	-48.3047
3.9833e-02	9.9031e+04	1.0702e+05	1.4581e+05	-47.2210
3.1638e-02	1.1121e+05	1.1792e+05	1.6209e+05	-46.6783
2.5126e-02	1.2408e+05	1.3134e+05	1.8068e+05	-46.6301
1.9957e-02	1.3844e+05	1.4802e+05	2.0267e+05	-46.9150
1.5849e-02	1.5553e+05	1.6845e+05	2.2927e+05	-47.2837
1.2590e-02	1.7685e+05	1.9267e+05	2.6153e+05	-47.4509
1.0001e-02	2.0415e+05	2.2014e+05	3.0023e+05	-47.1579